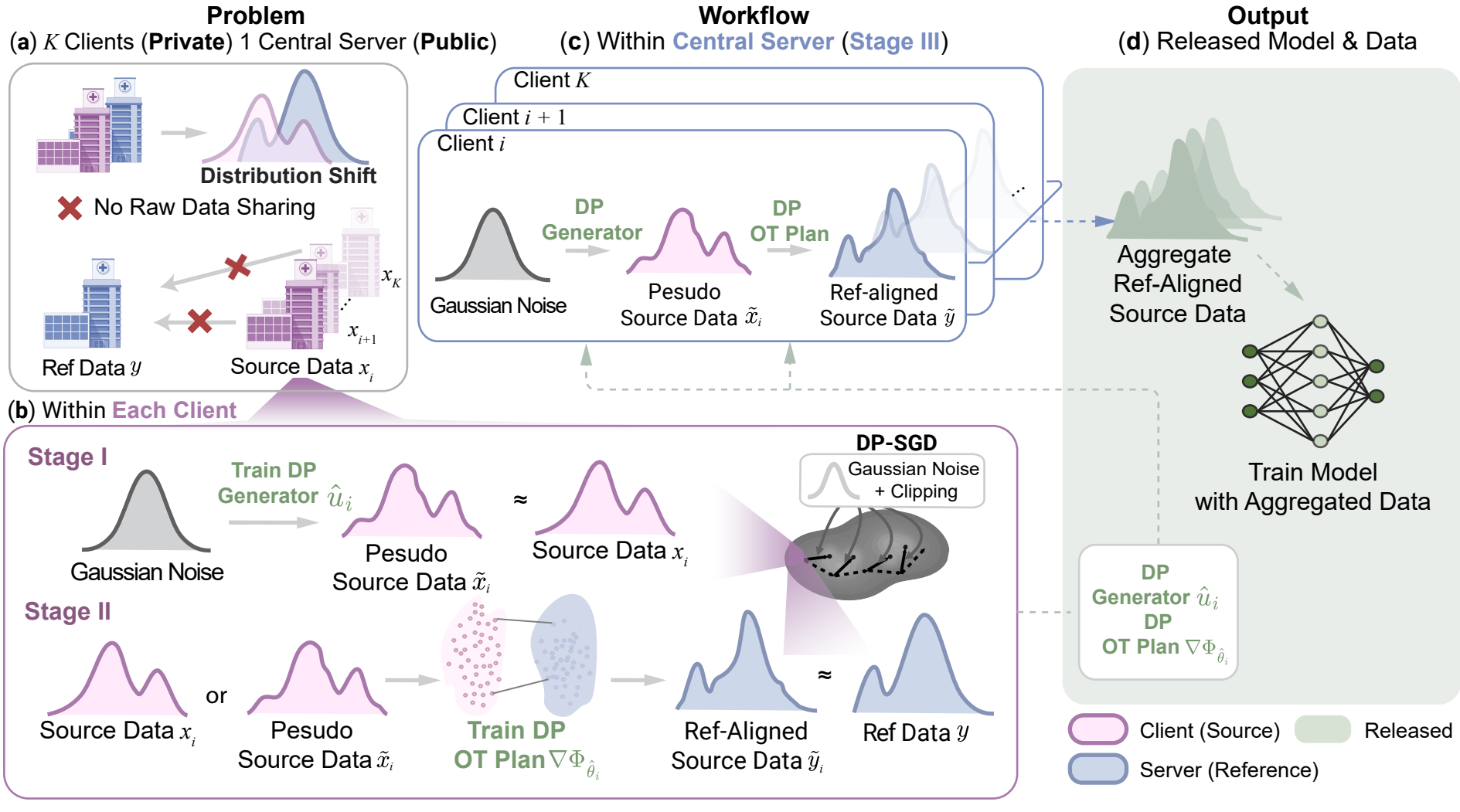
Problem
(a) K Clients (Private) 1 Central Server (Public)
Distribution Shift
No Raw Data Sharing
Ref Data y
Source Data x_i
x_K
x_{i+1}
Workflow
(c) Within Central Server (Stage III)
Client K
Client $i+1$
Client i
Gaussian Noise
DP Generator
Pesudo Source Data $\tilde{x}_i$
DP OT Plan
Ref-aligned Source Data $\tilde{y}$
Output
(d) Released Model & Data
Aggregate Ref-Aligned Source Data
Train Model with Aggregated Data
DP Generator $\hat{u}_i$
DP OT Plan $\nabla\Phi_{\hat{\theta}_i}$
(b) Within Each Client
Stage I
Gaussian Noise
Train DP Generator $\hat{u}_i$
Pesudo Source Data $\tilde{x}_i$
≈
Source Data x_i
DP-SGD
Gaussian Noise + Clipping
Stage II
Source Data x_i
or
Pesudo Source Data $\tilde{x}_i$
Train DP OT Plan $\nabla\Phi_{\hat{\theta}_i}$
Ref-Aligned Source Data $\tilde{y}_i$
≈
Ref Data y
Client (Source)
Released
Server (Reference)